

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024 batch	Date:	18/08/2026

Code of Conduct

I _____ C Prathibha _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ C Prathibha _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

ANSWERS

1. Constraint-Based Coreference Resolution

Given Paragraph

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedent(s)	Explanation
John	—	First introduction of entity
Mary	—	First introduction of entity
He	John, Mary	Gender constraint will eliminate Mary
a ball	—	New entity
She	Mary, John	Gender constraint will eliminate John
it	ball, John, Mary, park	Semantic compatibility strongly favors ball
The dog	—	New entity
him	John, Mary, dog	Gender and semantic constraints favor John
they	John and Mary, John and dog, Mary and dog	Coherence and context favor John and Mary
all	John, Mary, dog	Refers to all salient participants

Initial Mention List

Let:

- **E1 = John**
- **E2 = Mary**
- **E3 = park**
- **E4 = ball**
- **E5 = dog**

Pronoun variables:

- **H = He**
- **S = She**
- **I = it**
- **HM = him**
- **T = they**

b) Applying the Constraints

1. “He” → John

Possible antecedents: {John, Mary}

- **Gender agreement:** “He” requires a male antecedent.
- John is male.
- Mary is female.

Therefore:

He → John

2. “She” → Mary

Possible antecedents: {John, Mary}

- “She” requires a female antecedent.
- Mary is female.
- John is male.

Therefore:

She → Mary

3. “It” → Ball

Sentence:

“She wanted to play with it.”

Possible antecedents include {John, Mary, park, ball}.

Using semantic compatibility:

- “play with a ball” is natural.
- “play with John/Mary” is less suitable in this context.
- “play with the park” is not semantically appropriate.

Also, **ball** is the most recent suitable antecedent.

Therefore:

it → **ball**

4. “Him” → John

Sentence:

“The dog chased him excitedly.”

Possible antecedents: {John, Mary, dog}

Constraints:

- **Gender:** “him” normally refers to a male entity.
- John is male.
- Mary is female.
- The dog is the subject performing the chasing, so self-reference is not the preferred interpretation.
- Semantic compatibility supports a dog chasing a person.

Therefore:

him → **John**

5. “They” → John and Mary

Sentence:

“Finally, they all went home.”

Possible plural antecedents:

- {John, Mary}
- {John, dog}
- {Mary, dog}
- {John, Mary, dog}

The strongest interpretation is the group of previously discussed people, **John and Mary**. The word **all** can additionally include the dog in the event.

Thus:

they → **John and Mary**

and the final event can be understood as:

John, Mary, and the dog went home.

c) Constraint Graph / Constraint Table

Variables and Domains

Variable	Domain Before Constraints	Constraints Applied	Eliminated Links	Final Resolution
H = He	{John, Mary}	Gender, coherence	Mary	John
S = She	{John, Mary}	Gender, coherence	John	Mary
I = it	{John, Mary, park, ball}	Number, semantic compatibility, recency	John, Mary, park	ball
HM = him	{John, Mary, dog}	Gender, semantic compatibility, coherence	Mary, dog	John
T = they	{John+Mary, John+dog, Mary+dog, John+Mary+dog}	Number, coherence, discourse context	Other inconsistent groups	John+Mary
all	{John, Mary, dog}	Plurality, coherence	—	John+Mary+dog

Simple Constraint Graph

He

|

Gender ✓

|

John

She

|

Gender ✓

|

Mary

it

|

Semantic + Recency ✓

|

Ball

him

|

Gender + Semantic ✓

|

John

they

|

Number + Coherence ✓

|

John + Mary

d) Final Coreference Chains and Rewritten Paragraph

Coreference Chains

Chain 1: John

John → He → him

Chain 2: Mary

Mary → She

Chain 3: Ball

a ball → it

Chain 4: Group

John and Mary → they

Final event participants:

John + Mary + dog → all

Rewritten Paragraph with Clear References

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

e) Priority Order of Constraints

A suitable priority order is:

1. Gender and Number Agreement

This should be checked first because it quickly removes impossible antecedents.

Example:

“She” cannot refer to John.

2. Semantic Compatibility

After grammatical constraints, semantic meaning is important.

Example:

“play with it” strongly supports **ball**.

3. Coherence

The interpretation should maintain consistent entity chains throughout the discourse.

For example:

John → He → him

This creates a coherent chain.

4. Recency

Recency is useful when multiple candidates remain possible, but it should generally be lower than strong grammatical and semantic constraints.

What if Recency Is Relaxed?

If recency is relaxed, older entities can remain candidates for longer. This may sometimes be useful, but ambiguity increases.

For example, in:

“She wanted to play with it.”

Without recency, **John**, **Mary**, **park**, and **ball** may all initially remain possible. Semantic compatibility still selects **ball**, but the system must perform more reasoning.

Therefore, **recency improves efficiency and helps choose among equally compatible antecedents, but it should not override gender, number, or semantic constraints.**

2. Constraint-Based Dialog Generation

Conversation History

User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage

a) Relevant Discourse Planning Steps Before Generation

Before generating the response, the dialog system should perform the following steps:

Step 1: Identify the User's Situation

The important entities are:

- **you**
- **exam**
- **concentrate**

The user's problem is that the exam is tomorrow and the user cannot concentrate.

Step 2: Identify the Required Dialog Acts

The response must contain both:

1. **Advice** — provide a practical suggestion.
2. **Encouragement** — motivate the student positively.

Step 3: Maintain Entity Coherence

Reuse the important entities naturally:

- exam
- concentrate
- you

This keeps the response connected to the user's statement.

Step 4: Plan a Discourse Relation

Use a logical relation such as:

Cause → Effect

Example:

Taking a short break can help you focus better.

Or:

Elaboration

Example:

Try studying in short sessions, focusing on one topic at a time.

Step 5: Check Required Keywords

At least two of these must appear:

- focus
- break
- confident

Step 6: Check Final Constraints

Ensure:

- Only 2–3 sentences.
- Advice + encouragement are included.
- No contradictory information.
- Polite and positive tone.
- Entity coherence is maintained.

b) Three Possible Responses

Response 1

Take a short **break**, then return to your study and **focus** on one topic at a time so you can concentrate better for your **exam**. You have prepared for this, so be **confident** in yourself—you can do well tomorrow.

Response 2

If you are unable to **concentrate**, take a 10-minute **break** because a relaxed mind can help you **focus** better on your **exam** preparation. Stay **confident**, and remember that every small effort you make now can help you tomorrow.

Response 3

Try to **focus** on the most important topics for your **exam**, and take a short **break** whenever you feel mentally tired so you can concentrate again. You are capable, so stay **confident** and do your best.

c) Evaluation and Best Response

Criteria	Response 1	Response 2	Response 3	
Advice included	✓	✓	✓	
Encouragement included	✓	✓	✓	
Reuses exam/concentrate/you	Mostly	✓	✓	
At least 2 keywords	✓	✓	✓	
Cause-effect relation	Partial	✓ Strong	✓	
Positive tone	✓	✓	✓	
2–3 sentences	✓	✓	✓	
Logical consistency	✓	✓	✓	

Best Response: Response 2

Response 2 is the best choice.

It explicitly maintains the important entity **concentrate**, directly refers to the **exam**, and addresses **you** through the advice. It also has a clear **cause-effect discourse relation**:

taking a break → relaxed mind → better focus

It satisfies all required keyword, coherence, politeness, encouragement, and length constraints.

d) Effect of Violating Two Constraints

1. Violating Entity Coherence

Suppose the response suddenly talks about an unrelated topic:

“Don't worry about your exam. Playing football is always enjoyable.”

This would make the response less relevant because it does not maintain the entities **exam** and **concentrate**. The user may feel that the system did not understand the problem.

2. Violating Logical Consistency

Suppose the response says:

“Take a break and don't study at all, but make sure you study continuously for the whole night.”

This is contradictory. Such inconsistency reduces trust and makes the advice confusing.

Therefore, violating entity coherence reduces **relevance**, while violating logical consistency reduces **clarity and reliability**.

3. Word Sense Disambiguation, Logic and Discourse Structure

Source Sentence

“The bank by the river flooded after the storm, but it was saved by quick action.”

The word **bank** is ambiguous between:

1. A financial institution.
2. The land along the side of a river (**riverbank**).

a) Word Sense Disambiguation

Correct Sense: Riverbank

The correct interpretation is:

bank = riverbank

Justification

The phrase:

“by the river”

strongly provides geographical context.

The predicate:

“flooded after the storm”

is also semantically compatible with a riverbank. A storm can cause the river to overflow and flood its bank.

The pronoun:

“it”

also maintains coherence by referring back to the riverbank.

Constraint-Based Decision

Candidate Meaning	Context Compatibility	Semantic Compatibility	Final Result
Financial institution	Low	A bank building can flood, but “by the river” is less direct for this sense	Rejected
Riverbank	Very High	Strongly compatible with river and flooding	Selected

Thus:

bank → riverbank

b) Predicate Logic Representation

Let:

- **b** = the riverbank
- **r** = the river
- **s** = the storm
- **a** = quick action

Predicates:

- Riverbank(x) = x is a riverbank
- River(x) = x is a river
- Location(x,y) = x is located by y
- Storm(x) = x is a storm
- After(x,y) = x occurred after y
- Flood(x) = x flooded
- QuickAction(x) = x is quick action
- SavedBy(x,y) = x was saved by y

Predicate Logic

Riverbank(b)

$\wedge$ River(r)

$\wedge$ Location(b, r)

$\wedge$ Storm(s)

$\wedge$ After(Flood(b), s)

$\wedge$ Flood(b)

$\wedge$ QuickAction(a)

$\wedge$ SavedBy(b, a)

$\wedge \text{Contrast}(\text{Flood}(b), \text{SavedBy}(b, a))$

A simplified logical representation is:

$\text{Riverbank}(b) \wedge \text{River}(r) \wedge \text{Location}(b, r)$

$\wedge \text{Storm}(s)$

$\wedge \text{Flood}(b)$

$\wedge \text{After}(\text{Flood}(b), s)$

$\wedge \text{SavedBy}(b, a)$

$\wedge \text{QuickAction}(a)$

c) Target Language Sentence

Simple English Paraphrase

The riverbank beside the river was flooded after the storm, but quick action saved the riverbank.

This satisfies the constraints because:

- The ambiguity of **bank** is removed.
- Flooding is preserved.
- The storm is preserved.
- Quick action is preserved.
- The **contrast relation** is maintained using **but**.
- The pronoun ambiguity is removed, improving coherence.

Tamil Translation

புயலுக்குப் பிறகு ஆற்றங்கரை வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவான நடவடிக்கையால் அது பாதுகாக்கப்பட்டது.

d) Simple RST-Style Discourse Tree

The sentence has two main clauses:

1. **The riverbank flooded after the storm**
2. **It was saved by quick action**

These are connected by the relation **Contrast**.

CONTRAST

/ \

/ \

Clause 1

Clause 2

The riverbank flooded	It was saved
after the storm	by quick action

Another representation:

[CONTRAST]	
/	\
[Situation 1]	[Situation 2]
Flood(riverbank)	SavedBy(riverbank,
After(storm)	quick_action)

The word **but** signals that although the riverbank was flooded, the negative situation was followed by a positive outcome.

e) Advantages of Constraint-Based Approach Over Pure Statistical Machine Translation

1. Better Word Sense Disambiguation

A constraint-based approach uses contextual rules and semantic restrictions.

Here:

bank + river + flooded

strongly supports:

riverbank

A purely statistical system may choose the wrong meaning if the financial meaning of “bank” is more frequent in its training data.

2. Better Coreference Resolution

The system can explicitly resolve:

“it” → riverbank

This avoids losing the connection between the two clauses.

3. Preserves Semantic Relations

Constraints ensure that important information is retained:

- riverbank

- river
- storm
- flooding
- quick action
- saving

Thus, there is less information loss.

4. Maintains Discourse Structure

The system explicitly preserves:

Contrast

between:

flooding

and

being saved.

A statistical translation system may produce grammatically correct words but weaken or omit the rhetorical relation.

5. More Explainable

A constraint-based approach can explain its decision:

“Bank” was selected as “riverbank” because of the context “by the river” and the event “flooded after the storm.”

This makes the result transparent and easier to verify.

Conclusion

A **constraint-based approach** is particularly useful for sentences containing ambiguity, pronouns, and important discourse relations. For this sentence, it helps correctly identify **bank as riverbank**, resolve **it as riverbank**, preserve the flooding and saving events, and maintain the **contrast** relationship between the two clauses.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____